

IBM Data Science Module 8 Unit 3 Live session tutor guide

Session overview

This session is an opportunity for learners to engage with each other and the teaching team in real-time about the content covered in the module. Following a short introduction, the teaching team will demonstrate how to complete the steps of a specific activity before addressing questions from the cohort.

Hosted by: Head Tutor

Session length: 90 minutes

Purpose: To demonstrate how to build an end-to-end ML pipeline, including data preparation, training, tuning, and evaluation

Format: Step-by-step walkthrough of the process of building an end-to-end ML pipeline, including data preparation, training, tuning, and evaluation using the scenario and historical rainfall data from the 8.8 Activity.

Outcomes: Learners are able to explain the different evaluation techniques for supervised and unsupervised ML models, evaluate the performance of classification models on real-world data, and review key strategies for model validation and common pitfalls associated with modeling

Preparation

Regarding the demonstration, the host should:

- Ensure they have access to the activities on the Skills Network, specifically the 8.8 Activity: Building a rainfall prediction classifier
- Run the virtual environment setup
- Pre-test all the commands and activities in the environment

Before the session, the host should consult the live session announcement thread where learners will post their questions to prepare.

The host should meet with the technical support specialist beforehand to discuss the following ahead of the session:

- Clarity on everyone's role during the live session. For example, who will monitor the chat and relay questions or comments to the host?
- Host's ease with using Zoom
- Expected sequence of events
- Using the live session slide deck

- If questions are gathered from learners on the Online Campus beforehand, who will share the most-liked questions with the host, and when?
- Outline for learner engagement (outlined in the section below)
- The technical support specialist should be provided with the link for any assignment submission discussion so that they can post it if any project-related questions arise.

Recommendation for learner interaction

Learners are muted and chat through text. A technical support specialist (or a co-host) monitors the chat and selects questions or prompts for the live session host.

Outline for the live session

The host can use the information below, in combination with the live session slide deck and their own course notes, to facilitate the live session.

Waiting room (5 minutes before start)

Learners can use this window of time to familiarize themselves with the live session interface before the session begins.

Welcome and introduction from host (2 minutes)

The first slide of the live session presentation should be shared with learners from the time they are admitted from the waiting room. The host should briefly introduce themselves, their role on the teaching team, and their role in the technology space, or experience working on the course.

Optional: Live session etiquette (3 minutes)

The host should let learners know that they will all be muted. Depending on the structure of the live session, the host may also choose to highlight some items from the live session etiquette guide that are most relevant. It is especially important to explain to learners that the approach for student engagement is the following: Learners are muted and chat through the text chat feature. A technical support specialist (or a co-host) monitors the chat and selects questions or prompts for the live session host.

Some suggested phrases to use include:

- Feel free to post any questions you have in the chat box.
- When you ask a question, please be mindful of the time constraints that we are working around. Be sure to phrase your question thoughtfully and succinctly.
- Similarly, if you ask multiple questions, please be patient, as we will also try to answer questions from different learners so that everyone has an opportunity to participate in the session.
- This session presents a great opportunity for us all to connect and discuss the course content. The Tutor Team makes sure that questions in the discussions are answered, and, in that way, you will have a written record of the answer to your

question. Our technical support assistant will post the link to that discussion in this chat so that you can easily navigate to it.

- This session is being recorded and will be posted on the Online Campus for your later access, should you need it.

Agenda (2 minutes)

Briefly outline the agenda for the session.

- Introduction and live session etiquette (5 minutes)
- Agenda (2 minutes)
- Live demonstration (45 minutes)
- Session consolidation (5 minutes)
- Looking ahead (3 minutes)
- Office hours (30 minutes)
 - Questions submitted prior to the session on the discussion
 - Questions posed during the live session

Total run time: 90 minutes

Live demonstration (45 minutes)

This session should include a demonstration of the work learners have engaged with throughout the module. Any questions that fall within the scope and depth of the live session topic and course content so far should be addressed during this time. If learners have additional questions during the live session chat, these can also be discussed, where relevant and appropriate.

State the lesson's purpose

In this live session, you'll have a final opportunity to consolidate all core concepts from the module and reinforce key distinctions and practical strategies for ML implementation.

Step 1: Explore and prepare the dataset. This includes feature engineering and cleaning.

Step 2: Build a classifier pipeline. This includes model selection, training, and optimization.

Step 3: Evaluate the model's performance. This includes interpreting metrics and visualizations.

Session consolidation (5 minutes)

The host will use this time to consolidate the session, highlighting any key themes that learners should keep in mind during their learning journey.

Looking ahead (3 minutes)

The closing portion of the live session will prompt learners to reflect on how the content covered in the session is applied or utilized in the real world. Learners should be encouraged to post their reflections in the chat box.

Office hours (30 minutes)

The host will then inform the learners that they are all welcome to take the next 30 minutes to discuss difficulties related to building an end-to-end ML model, and any other areas covered in the live session or module, address unanswered questions, and network with each other. This is an opportunity to connect, share ideas, or dive deeper into any points that stood out to them. The learners should be encouraged to use the time in whatever way best suits their needs.